

Volume Control Transfers, Conditioning Does Not: The Locality of Implicit EM in Neural Networks

Abstract

Log-sum-exp objectives produce responsibility gradients: for an objective over distances or energies, the derivative with respect to each distance coordinate is the posterior responsibility of the corresponding component. This identity explains how standard neural objectives can instantiate generalized EM without explicit latent-variable updates. We show that the identity is local. In distance coordinates, the LSE Hessian is uniformly bounded by $1/2$; with private affine component parameters, this bound is parameter-independent. When the same site is composed behind a learned dense map, the leading term of the composed cross-entropy Hessian is a conjugated softmax Hessian whose bound scales with the spectrum of the learned map, and the parameter-independent guarantee no longer holds.

Experiments on supervised MNIST networks show that the two consequences of this analysis dissociate under composition: volume control transfers, while conditioning does not. Applying volume control at an intermediate EM site eliminates dead units and reduces redundancy across widths, indicating that local mixture failures can be corrected within a supervised model. The optimizer behavior characteristic of the single-layer implicit-EM model, however, does not survive supervised composition: SGD regains its learning-rate sensitivity and adaptive methods recover their advantage. Gradient diagnostics at the EM site exhibit the same pattern. At the standard auxiliary weight, the corridor dominates the weighted local EM path by roughly $67\text{--}95\times$ late in training, with no detectable alignment or opposition between the two directions. Blocking the corridor with a stop-gradient restores learning-rate insensitivity; increasing the EM weight until the EM path dominates produces the same behavior without modifying the computation graph. Implicit EM is therefore local and graded: volume control corrects failures at EM sites, but EM conditioning appears only when the responsibility-gradient path directly or dominantly controls the effective update, and simplex-compatible transport remains a theoretical first-order preservation class.

1 Introduction

Prior work showed that gradient descent on log-sum-exp distance objectives implements implicit expectation-maximization in a precise local sense: the gradient with respect to each component distance is the component responsibility [23]. A decoder-free single-layer encoder tested the unsupervised consequences of that identity and confirmed the predicted local behavior [24].

That study also exposed a more restrictive property. The single-layer model had optimizer dynamics that ordinary neural networks do not generally exhibit: SGD was insensitive to learning rate over a wide range, Adam offered little representational advantage, lower loss did not necessarily imply better features, and convergence time was roughly fixed. These behaviors are EM-like in a stronger sense than the gradient identity alone. They suggest that the local responsibility structure can condition the effective parameter update.

Those optimizer dynamics do not hold for neural networks in general, even though many standard objectives contain log-sum-exp or softmax structure. Cross-entropy classifiers, attention layers,

contrastive objectives, and energy-based models all contain exponentiation-and-normalization interfaces, but they do not usually train with the learning-rate insensitivity of the single-site implicit-EM model. The question is therefore not whether responsibility gradients exist. The question is where, under composition, the optimizer consequences of those gradients are lost.

This paper studies that question in the supervised regime. The testbed is a minimal supervised network with an explicit intermediate distance-to-responsibility site. The first layer produces component distances, a NegLogSoftmin transform creates a calibrated EM interface, and a learned dense map W_2 sends those component coordinates to class logits. A local auxiliary objective gives the intermediate site the same volume-control machinery used in the decoder-free model, while the supervised cross-entropy objective pulls gradients back through the dense head.

The outcome separates two EM properties. Volume control transfers: when applied at the intermediate site, it eliminates dead units and reduces redundancy across widths. The optimizer signature does not transfer under ordinary supervised composition: SGD becomes learning-rate sensitive again, and adaptive methods recover their advantage. The network contains a valid responsibility-gradient site, but the deeper supervised model no longer behaves like the isolated implicit-EM model.

The mechanism is ownership of the update. The local identity creates an EM-structured path at the intermediate site, but the optimizer behavior at W_1 depends on whether that path controls the effective update. In the supervised model, the cross-entropy gradient reaching W_1 is pulled back through W_2 and added to the local EM gradient. When that supervised corridor dominates, the update at W_1 is governed by a composed, non-EM contribution even though the local responsibility site still exists.

The resulting boundary has three parts. Implicit EM is *local*: the responsibility structure exists at LSE and softmax sites, and it extends to parameters only along paths that preserve component identity. It is *graded*: the relevant variable is the measured share of EM-structured gradient in the effective update, not merely whether the computation graph is connected. It is *structural*: at-site failures such as dead or redundant components are different from corridor failures caused by learned dense composition.

The theory formalizes this boundary. In distance coordinates, the LSE Hessian has spectral norm at most $1/2$. With private affine component parameters, this gives a parameter-independent curvature bound. When the same site is trained through a learned dense supervised corridor, the leading Gauss–Newton term of the cross-entropy Hessian is a conjugated softmax Hessian whose scale depends on the learned map. The local LSE guarantee no longer controls the composed update.

The experiments show the same split. Volume-control ablations confirm that local mixture failures can be repaired inside a supervised model. Optimizer sweeps show that the single-site signature disappears under ordinary supervised composition. A causal intervention identifies the mechanism: blocking the CE corridor to W_1 with stop-gradient restores learning-rate insensitivity, and increasing the EM weight until the local path dominates restores the same behavior without cutting the graph. Gradient diagnostics match the intervention. At the standard auxiliary weight, the CE corridor dominates the weighted local EM path by roughly $67\text{--}95\times$ late in training, with no detectable alignment or opposition between the two directions.

This paper makes the following contributions.

1. It identifies the missing condition for EM-like optimizer dynamics in composed networks: the responsibility-gradient path must directly or dominantly control the relevant parameter update.
2. It proves a conditioning contrast between private-affine LSE objectives and LSE sites com-

posed behind learned dense maps, and identifies operations that preserve or destroy the responsibility-gradient invariant.

3. It separates at-site failures from structural corridor failures in supervised networks with intermediate EM sites.
4. It shows that volume control transfers to intermediate supervised EM sites, repairing dead or redundant components, while the single-site optimizer signature does not survive ordinary dense supervised composition.
5. It restores the conditioning signature by stop-gradient and by EM dominance, and uses the $1/2$ -saturation diagnostic to measure local mixture sharpness at the EM site.

2 Related Work and Positioning

This paper asks why the EM-like optimizer dynamics observed in an isolated implicit-EM model do not generally appear in composed neural networks, even when those networks contain log-sum-exp or softmax interfaces. The relevant prior work therefore falls into three groups: EM as an optimization method, neural mechanisms that produce responsibility-shaped gradients, and training methods that protect or balance local objectives under composition.

2.1 EM as Optimization and Conditioning

The expectation-maximization algorithm was introduced for maximum-likelihood estimation with incomplete data [8]. Its classical form alternates between computing posterior responsibilities and maximizing an expected complete-data objective. The free-energy view of EM [20] makes this coordinate-ascent structure explicit and clarifies why EM can be studied as an optimization procedure rather than only as a probabilistic modeling algorithm.

The closest optimization precedent is the analysis of EM as a preconditioned gradient method [33]. That line of work treats EM’s behavior as a consequence of responsibility geometry and missing information. This matters for the present paper because the decoder-free implicit-EM model exhibited optimizer dynamics that are unusual for neural networks: SGD was learning-rate insensitive over a wide range and adaptive optimization offered little representational advantage. The question here is when a neural responsibility-gradient site inherits enough of EM’s optimization geometry to produce those dynamics.

The local curvature constant used in our theory is not itself new. The softmax or multinomial-logistic Hessian bound follows from the classical bound of Bohning [6], and can also be obtained from an elementary variance argument. The contribution here is to locate where that bound applies in a neural objective and where it stops applying after learned composition.

2.2 Responsibility Gradients in Neural Networks

Several neural architectures and objectives already contain responsibility-like quantities. Mixtures of experts use learned soft gates over expert heads and admit EM-style training [14]. Attention mechanisms use softmax routing over values [32], and associative-memory interpretations make the retrieval geometry explicit [28]. More recently, Agarwal et al. [1] derived first-order attention training dynamics in which cross-entropy induces advantage-form attention-score updates and value updates weighted by attention responsibilities. They frame this training-time correspondence as EM-like and structural rather than variational: the derived updates have the form of

responsibility-weighted and advantage-based EM-like steps. The implicit-EM account [23] sharpens this interpretation. In the distance-based LSE case, when neural outputs are read as distances or energies, the log-sum-exp term is a mixture marginal whose gradients are posterior responsibilities. Under that interpretation, the responsibility-gradient algebra is not merely EM-like; it is implicit EM.

This gives a more general local statement for log-sum-exp distance objectives: the gradient with respect to each component distance is its posterior responsibility. It places unsupervised mixture-like objectives, supervised cross-entropy, and attention-style softmax interfaces under a shared local algebra. The present paper does not address inference-time Bayesian computation in trained attention models. It asks the next training-time question: when such responsibility-shaped gradients exist locally, do their optimizer consequences survive the pullback through the rest of the network?

2.3 The Single-Site Baseline

The decoder-free sparse-autoencoder study tested the unsupervised branch of the implicit-EM theory [24]. A single encoder with an LSE objective and volume control learned structured prototype-like features. The expected local failures appeared under ablation: without volume control, components collapsed or died; with only anti-collapse control, components remained redundant; with both variance and decorrelation, the site remained active and diverse.

The more restrictive observation was the learning dynamics. In that isolated single-site setting, SGD was insensitive to learning rate over several orders of magnitude, Adam did not improve representation quality, and lower loss did not necessarily produce better features. These are the properties whose survival is tested here. The supervised model is not introduced merely to show that volume control can regularize a hidden layer. It is a test of whether the learning dynamics of the isolated implicit-EM site persist under supervised composition.

2.4 Volume Control and Collapse

Classical mixture models motivate the distinction between local mixture health and optimizer conditioning. Mixture likelihoods can develop singular components, nearly hard assignments can leave components with little chance of recovery, and diffuse responsibilities can slow specialization [19, 29]. In the neural setting, the corresponding at-site failures are dead components, collapsed variance, redundant features, and weakly calibrated responsibilities.

The variance and decorrelation terms used here play the local role of volume control. They are also related to redundancy-reduction methods in self-supervised learning, including Barlow Twins and VICReg [4, 35]. Those methods were developed for representation learning under invariance objectives. Here the same structural motif is attached to an intermediate EM site. One contribution of the present paper is to show that this local repair transfers into a supervised network even when the optimizer signature of the isolated site does not.

2.5 Composition, Local Learning, and Credit Assignment

The failure mode studied here is compositional. A local EM site may create a responsibility-gradient path, but ordinary backpropagation also pulls foreign gradients through learned maps before they reach earlier parameters. This places the work near methods that protect or decouple local objectives.

Deep belief networks and greedy layer-wise pretraining showed that useful representations can be learned by protected local objectives [12]. Later work on greedy layer-wise learning [5], decoupled

neural interfaces and synthetic gradients [13], and Forward-Forward [11] also explores alternatives to fully end-to-end credit assignment. Auxiliary classifiers and deeply supervised networks provide a supervised route for injecting intermediate losses into deep models [17, 31].

Our contribution is not a new local-learning algorithm. It is a mechanism for why a protected local objective can preserve optimizer structure that ordinary dense backpropagation may obscure. In the implicit-EM setting, the protected object is the responsibility-gradient path. If it reaches private component parameters directly, the local LSE bound controls the update. If the same parameters are trained mainly through a dense supervised corridor, the local bound no longer governs the effective dynamics.

2.6 Auxiliary Losses, Dominance, and Interference

Training with both a main CE loss and a local auxiliary EM loss places the work near auxiliary-loss and multi-task learning. PCGrad [34] modifies conflicting task gradients, GradNorm [7] balances task weights, and gradient starvation [26] studies one signal dominating another. These methods address practical optimization conflicts among multiple training signals.

The mechanism measured here is dominance rather than direct opposition. The cosine between the CE and local-EM gradients at the intermediate distance coordinates is near zero, not strongly negative. The EM path is diluted by a larger foreign path rather than cancelled by an opposing one. This is why the gradient ratio, not a conflict metric, is the central diagnostic for whether the EM-like optimizer signature is visible at W_1 .

2.7 Generative and Discriminative Training

The generative–discriminative distinction [21] is a natural high-level explanation for why the isolated LSE model and the supervised classifier behave differently. The present result is compatible with that distinction but more specific. In the causal experiment, the data, architecture, labels, and local site are held fixed while the route by which cross-entropy reaches W_1 is changed. Cutting the CE corridor restores the single-site optimizer signature; increasing the EM weight until the local path dominates restores the same behavior without cutting the graph.

Thus the claim is not simply that generative and discriminative objectives differ. The paper identifies the corridor through which the discriminative objective reaches the local EM site and shows when that corridor controls the effective update.

2.8 Loss Landscapes and Basin Structure

The loss-landscape literature provides useful future diagnostics for this setting. Linear mode connectivity shows that networks trained in apparently different basins can often be connected by low-loss paths [10]. Permutation invariance is a central complication: component or neuron re-labeling can create artificial barriers unless models are aligned before interpolation [2, 9]. Linear probes are a first diagnostic for representation alignment [3].

These tools are relevant because EM-dominated and CE-dominated regimes can produce representations with different task alignment and different local geometry. A basin analysis would ask whether discriminative fine-tuning refines the EM solution or moves to a distinct region after hidden-unit alignment. That analysis is outside the present evidence. The central evidence here is the stop-gradient and dominance intervention.

2.9 Scope and Positioning

This paper continues a specific line of inquiry. The attention-gradient work of Agarwal et al. [1] showed that cross-entropy training of attention has EM-like local dynamics, while explicitly treating the training-time EM correspondence as structural. The implicit-EM identity then identified the exact local algebra behind responsibility gradients in distance-based LSE objectives. The decoder-free study [24] tested the unsupervised consequences of that local identity and revealed the optimizer signature that motivates the present paper.

The current contribution is a boundary result. It is not a general theory of deep-network optimization, Hessian spectra, or generalization. It concerns log-sum-exp and softmax sites, the responsibility gradients those sites create, and the corridors through which those gradients reach parameters. Within that scope, the paper separates three properties that can otherwise be conflated: existence of a local responsibility-gradient interface, local volume-controlled mixture health, and EM-like optimizer dynamics at the parameters.

3 Background: The Locality of Implicit EM

This section fixes the conceptual object studied in the rest of the paper. An implicit-EM site is not any hidden layer with a softmax nearby, and it is not an entire deep network. It is a local exponentiation-and-normalization interface at which an objective sees distances, energies, or logits and therefore produces a responsibility-shaped gradient. The theory section asks when this local structure survives the pullback to parameters.

3.1 Responsibilities as Gradients

Log-sum-exp objectives are ubiquitous in neural networks. Softmax classification, attention, contrastive losses, and energy-based objectives all contain the same exponentiation-and-normalization geometry [1, 16, 22, 32]. In distance or energy coordinates, that geometry has a simple consequence: gradients are responsibilities. If

$$L_{\text{LSE}}(d) = -\log \sum_j \exp(-d_j), \quad r = \text{softmax}(-d),$$

then

$$\frac{\partial L_{\text{LSE}}}{\partial d_j} = r_j.$$

The posterior responsibility of component j is not an auxiliary variable computed outside the optimizer. It is the gradient supplied to gradient descent.

This identity gives a precise sense in which standard objectives can implement implicit generalized EM. The E-step appears as a normalized gradient over components; the parameter update plays the role of a generalized M-step [8, 20, 33]. The identity is purely algebraic; it assumes no particular parameterization, architecture, or optimizer.

Everything in this paper turns on what happens next: the responsibility vector may reach private component parameters directly, or it may be mixed with other gradients and pulled through learned maps before it updates earlier weights.

3.2 Cross-Entropy as a Label-Clamped Responsibility Interface

Supervised cross-entropy has the same exponentiation-and-normalization interface, with the label clamping one component: its logit gradient is $p_c - \mathbb{I}\{c = y\}$ (Appendix A), removing probability mass from the true class and redistributing it according to the model posterior.

This observation supports a limited conclusion and invites an incorrect one. It is correct that the output loss supplies responsibility-shaped geometry at the output logits. It is not correct to conclude that every earlier hidden layer is itself performing EM. The gradient reaching an earlier layer has already passed through the head Jacobian and any intervening nonlinearities; it need not be nonnegative, sum to one, or remain aligned with hidden components.

3.3 Local EM Sites Inside a Network

The supervised architecture studied here deliberately creates an intermediate site. The first map produces component distances $d_1, \dots, d_K$, and the NegLogSoftmin transform (Appendix A) calibrates them so that $\exp(-\text{NLS}(d)_j) = r_j$, giving the hidden layer an explicit distance-to-responsibility interface rather than relying on the output loss to impose structure indirectly.

The distinction between an output interface and an intermediate site is central. At the output, the supervised objective owns the softmax. At the hidden layer, the local site exists only because we add a local auxiliary objective. Without such an objective, the hidden coordinates may be useful features, but there is no reason for their updates to be posterior responsibilities over hidden components.

3.4 Volume Control

An LSE competition term by itself does not prevent the standard degeneracies of mixture models. Components can collapse, become redundant, or receive essentially no responsibility [19, 29]. The decoder-free encoder study addressed these failures with volume control: a variance term prevents dead or collapsed components, and a decorrelation term prevents all components from moving together. In the local auxiliary objective $L_{\text{aux}} = L_{\text{LSE}} + L_{\text{var}} + L_{\text{tc}}$ (Appendix A), the variance term is the anti-collapse part and the decorrelation term the anti-redundancy part. This variance-plus-decorrelation pairing is structurally close to redundancy-reduction regularizers such as VICReg and Barlow Twins, although here it serves a mixture-geometry role at an EM site rather than a self-supervised invariance objective [4, 35]. Together the terms act as a volume barrier, encouraging components to occupy distinct directions rather than concentrate in the same small region. The role is analogous to that of volume terms in Gaussian mixture likelihoods; we do not claim an identification with the Gaussian log-determinant.

The decoder-free encoder study also exhibited an unusual optimizer signature: SGD was insensitive to learning rate over a wide range, Adam offered little representational advantage, lower loss did not necessarily imply better features, and convergence time was roughly fixed [24]. This optimizer behavior is the phenomenon whose transfer fails in the supervised setting studied here.

The first empirical question is whether these local repairs still work inside a supervised network. They do: Section 6 shows that volume control eliminates dead units and reduces redundancy at the intermediate site. The second question is whether the optimizer conditioning observed in the single-layer implicit-EM model transfers through the supervised head. Sections 7 and 8 show that it does not, except when the local EM path is isolated or made dominant.

3.5 Locality Versus Inheritance

The main conceptual error this paper rules out is inheritance. The gradient-responsibility identity guarantees a local structure at the coordinates seen by an LSE or softmax objective. It does not guarantee that the whole network inherits EM’s conditioning. Backpropagation transports the gradient, but transport does not preserve its structure. Dense learned maps can mix signs and components; objective sums can add unrelated gradients; gates can turn small responsibilities into exactly zero updates.

These are the failure operations formalized in the next section: sign-indefinite mixing, foreign objective sums, and hard gates break the private component structure that makes the local responsibility update well conditioned.

There are accordingly two claims one might make about a hidden layer. A weak, correct claim is that loss geometry at downstream softmax sites shapes hidden representations through backpropagation. A strong, generally false claim is that the hidden layer itself receives a posterior responsibility update. The rest of the paper draws the boundary between these claims.

3.6 Two Failure Classes

The experiments are organized around two failure classes. At-site failures occur at a local EM interface: dead components, collapsed variance, redundant features, and diffuse or poorly calibrated responsibilities. Volume control is designed to repair the first three by keeping components alive and distinct. Diffuse responsibilities are diagnosed separately by the 1/2-saturation measure introduced in Lemma 1: low curvature relative to the 1/2 bound indicates weak local competition rather than sharp two-way component assignment.

Structural failures occur when a responsibility gradient passes through a shared learned corridor before reaching the parameters it is supposed to condition. In the supervised model, this corridor is the component-to-class map W_2 . The CE gradient reaching W_1 is pulled through W_2 and combined with the local auxiliary gradient. When this foreign path dominates, the update at W_1 no longer has the private, per-component structure of an EM M-step.

The structural issue is the neural analogue of a classical EM fact, not a reduction of supervised cross-entropy to an EM complete-data objective. EM admits per-component M-steps exactly when the expected complete-data objective separates over component-private parameters [8, 19]. When components share parameters, the M-step becomes coupled and the per-component update is lost. Backpropagation through a learned dense corridor is parameter sharing in this structural sense. It creates a foreign contribution at the earlier parameters whose curvature is no longer controlled by the local LSE bound.

This structural claim must be separated from dominance. Sharing explains why the cross-entropy contribution is not EM-conditioned; dominance explains when that contribution controls the update. The gradient at W_1 is a sum,

$$\nabla_{W_1} L_{\text{CE}} + \lambda \nabla_{W_1} L_{\text{aux}}.$$

If the CE term dominates, the effective landscape is the composed supervised landscape. If the local auxiliary term dominates, the same graph can recover the single-site optimizer signature.

This split sets up the theory. The next section proves that private LSE sites inherit a parameter-independent curvature bound, while composition through a learned dense map loses that guarantee. The experiments then show the same boundary from both sides: local volume control transfers, but optimizer conditioning does not survive ordinary composition.

4 Theory: The Conditioning Boundary

The gradient-responsibility identity is local: it is a statement about the coordinates d at an exponentiation-and-normalization site. The question for a neural network is what happens after this responsibility vector is pulled back to parameters. This section isolates the condition under which the local curvature bound survives that pullback, and the condition under which it does not.

4.1 The Simplex Invariant

Lemma 1 (Simplex invariant and curvature bound). *For*

$$L_{\text{LSE}}(d) = -\log \sum_{j=1}^K \exp(-d_j), \quad r = \text{softmax}(-d),$$

the gradient $\nabla_d L_{\text{LSE}} = r$ lies on the probability simplex. The Hessian is

$$\nabla_d^2 L_{\text{LSE}} = rr^\top - \text{diag}(r).$$

Equivalently, the negative Hessian is the categorical covariance matrix

$$C(r) = \text{diag}(r) - rr^\top \succeq 0, \quad \|C(r)\|_2 = \|\nabla_d^2 L_{\text{LSE}}\|_2 \leq \frac{1}{2}.$$

This is the multinomial-logistic Hessian bound of Bohning [6].

Proof. The gradient calculation gives

$$\frac{\partial L_{\text{LSE}}}{\partial d_j} = \frac{\exp(-d_j)}{\sum_k \exp(-d_k)} = r_j.$$

Differentiating r with respect to d gives

$$\frac{\partial r_j}{\partial d_k} = -r_j(\mathbb{I}\{j = k\} - r_k),$$

so $\nabla_d^2 L_{\text{LSE}} = rr^\top - \text{diag}(r)$. For any vector v ,

$$v^\top C(r)v = \sum_j r_j v_j^2 - \left(\sum_j r_j v_j \right)^2 = \text{Var}_{j \sim r}(v_j).$$

By Popoviciu's inequality [27],

$$\text{Var}_{j \sim r}(v_j) \leq \frac{1}{4}(\max_j v_j - \min_j v_j)^2.$$

For $\|v\|_2 = 1$, the squared range is at most 2, hence $v^\top C(r)v \leq 1/2$. Taking the supremum over unit vectors proves the bound. $\square$

The important feature is not boundedness alone but the coordinates in which the bound holds. In distance coordinates it is absolute: no parameter value, scale, or training time appears. The gradient is a normalized conditional expectation, and the curvature is the covariance of that conditional distribution.

Proximity to the bound also gives a diagnostic used in Section 8: curvature approaching $1/2$ means the local responsibility vector is approaching sharp two-way competition, while values well below $1/2$ indicate diffuse responsibilities.

4.2 Private Parameters Inherit the Bound

Proposition 1 (Conditioning under private affine energies). *Let each component energy be a private affine map*

$$d_j = w_j^\top x + b_j,$$

and let $\theta = (w_1, b_1, \dots, w_K, b_K)$. For a single example,

$$\|\nabla_\theta L_{\text{LSE}}\|_2 \leq \sqrt{\|x\|^2 + 1},$$

and

$$\|\nabla_\theta^2 L_{\text{LSE}}\|_2 \leq \frac{1}{2}(\|x\|^2 + 1).$$

With biases absorbed into an augmented input $\tilde{x} = (x, 1)$, this is the bound $\|\nabla_\theta^2 L_{\text{LSE}}\|_2 \leq \frac{1}{2}\|\tilde{x}\|^2$. Under batch-mean training,

$$\|\nabla_\theta^2 L_{\text{LSE}}\|_2 \leq \frac{1}{2}\mathbb{E}\|\tilde{x}\|^2,$$

a constant determined by the data and independent of the current parameters.

Proof. Let $\tilde{x} = (x, 1)$ absorb the bias and write $d_j = \theta_j^\top \tilde{x}$. Since d is linear in the private parameter blocks, there is no second-order term from the map $\theta \mapsto d$. The gradient block for component j is $r_j \tilde{x}$, so

$$\|\nabla_\theta L_{\text{LSE}}\|_2^2 = \sum_j r_j^2 \|\tilde{x}\|^2 \leq \|\tilde{x}\|^2.$$

The Hessian has blocks

$$(\nabla_\theta^2 L_{\text{LSE}})_{jk} = \left(r r^\top - \text{diag}(r) \right)_{jk} \tilde{x} \tilde{x}^\top,$$

or equivalently $-C(r) \otimes \tilde{x} \tilde{x}^\top$. The minus sign follows from the distance convention: $C(r)$ is the negative Hessian of L_{LSE} in d -coordinates. The spectral norm of a Kronecker product is the product of spectral norms, and Lemma 1 gives the factor $1/2$. $\square$

Proposition 1 is the formal version of the single-site conditioning claim for the LSE term. The LSE smoothness constant is fixed before training starts. Whatever step-size range is allowed by the data-dependent constant does not shrink as the parameters move, and there is no learned anisotropy in this term for an adaptive optimizer to correct. This is the mechanism behind the optimizer signature observed in the single-layer implicit-EM model, insofar as the LSE term drives the update.

The bound covers only the LSE contribution. The full auxiliary objective used experimentally also contains variance and decorrelation terms, whose curvature is not covered by Proposition 1. We therefore use the proposition as the local conditioning baseline: it identifies the parameter-independent part supplied by the responsibility site, while the experiments measure the behavior of the full auxiliary objective in the composed model. In particular, L_{LSE} alone is not a complete well-posed training objective; the volume-control terms are what prevent collapse and make the learned representation useful.

Proposition 2 (Smooth monotone kernels). *Let*

$$d_j = \phi(w_j^\top x + b_j)$$

with private component parameters, $\phi' \geq 0$, $|\phi'| \leq B_1$, and $|\phi''| \leq B_2$. Then the parameter-space curvature of $L_{\text{LSE}}(d(\theta))$ is bounded by a constant multiple of $\|\tilde{x}\|^2$ that depends only on B_1 and B_2 , not on the current parameters.

Proof. The monotonicity condition $\phi' \geq 0$ preserves the first-order component orientation: increasing a preactivation cannot reverse a component's distance coordinate. The curvature bound itself uses only the uniform derivative bounds $|\phi'| \leq B_1$ and $|\phi''| \leq B_2$. The chain rule gives two terms:

$$\nabla_{\theta}^2 L = J^\top \nabla_d^2 L J + \sum_j \frac{\partial L}{\partial d_j} \nabla_{\theta}^2 d_j.$$

The first term is bounded by Lemma 1 and $\|J\|^2 \leq B_1^2 \|\tilde{x}\|^2$. The second term is block diagonal across components; since $\partial L / \partial d_j = r_j$ and $\sum_j r_j = 1$, its norm is bounded by $B_2 \|\tilde{x}\|^2$. The total curvature is therefore bounded by $(B_1^2/2 + B_2) \|\tilde{x}\|^2$, with biases handled by the augmentation convention above. $\square$

Softplus is the main smooth case in this paper. ReLU requires separate treatment: its curvature bounds hold almost everywhere, but it introduces a zeroth-order absorbing-state pathology. When a unit is gated off for the whole batch, the parameter-gradient block through that unit is exactly zero, and the unit may not recover. That failure is not a violation of the curvature bound; it is the neural version of hard-assignment cluster death.

4.3 Composition Does Not Preserve the Uniform Bound

We now compose the same local site behind a learned dense map. Let

$$y = \text{NLS}(d), \quad h = W_2 y, \quad L_{\text{CE}} = -h_c + \log \sum_{\ell} \exp(h_{\ell}),$$

and $p = \text{softmax}(h)$. The Jacobian of $y = \text{NLS}(d)$ is

$$J = \frac{\partial y}{\partial d} = I - \mathbf{1} r^\top,$$

because $\partial \log \sum_k \exp(-d_k) / \partial d_j = -r_j$.

Proposition 3 (Composition does not preserve the uniform bound). *For the composed supervised corridor above, the Hessian of L_{CE} with respect to the intermediate distances d decomposes as*

$$\nabla_d^2 L_{\text{CE}} = \underbrace{J^\top W_2^\top \left(\text{diag}(p) - p p^\top \right) W_2 J}_{\text{Gauss-Newton term}} + \underbrace{\sum_l \frac{\partial L_{\text{CE}}}{\partial y_l} \nabla_d^2 y_l}_{\text{NLS second-order term}}.$$

The Gauss-Newton term is bounded by $\frac{K}{2} \sigma_{\max}(W_2)^2$, a quantity that depends on the learned spectrum of W_2 ; the uniform, parameter-independent bound of Lemma 1 therefore no longer controls the composed curvature. No parameter-independent bound is claimed for the NLS second-order term.

The decomposition is the standard generalized Gauss-Newton separation between the positive semidefinite output-curvature pullback and the remaining model-curvature terms [18, 30]; the contribution is the contrast with the isolated site, not the decomposition.

Proof. The label clamp $-h_c$ in L_{CE} is linear in h , so it contributes to the gradient $p - e_c$ but not to the second derivative. The CE Hessian with respect to logits is therefore the softmax Hessian

$$H_h L_{\text{CE}} = \text{diag}(p) - p p^\top,$$

independent of the label. Pulling this term back through the linear map $h = W_2 y$ and the NLS Jacobian $J = \partial y / \partial d$ gives the displayed Gauss–Newton term $J^\top W_2^\top (\text{diag}(p) - pp^\top) W_2 J$; the remaining terms come from the second derivatives of $y = \text{NLS}(d)$, weighted by the incoming gradient from the head. To bound the leading term we need the two conjugating norms. First, the softmax Hessian satisfies $\|\text{diag}(p) - pp^\top\|_2 \leq 1/2$ exactly as in Lemma 1 and Bohning [6], so

$$\|W_2^\top (\text{diag}(p) - pp^\top) W_2\|_2 \leq \frac{1}{2} \sigma_{\max}(W_2)^2.$$

Second, we bound the NLS Jacobian $J = I - \mathbf{1} r^\top$, which is not a contraction. We compute $\|J\|_2$ in closed form. Writing

$$J^\top J = I - \mathbf{1} r^\top - r \mathbf{1}^\top + K r r^\top = I + M, \quad M := -\mathbf{1} r^\top - r \mathbf{1}^\top + K r r^\top,$$

the perturbation M is supported on $\text{span}\{\mathbf{1}, r\}$: for any v orthogonal to both $\mathbf{1}$ and r we have $Mv = 0$, so $J^\top J$ acts as the identity on that complement (eigenvalue 1). If $r = \mathbf{1}/K$, then J is the orthogonal projector onto $\mathbf{1}^\perp$, so the eigenvalues of $J^\top J$ are 1 with multiplicity $K - 1$ and 0 once. Otherwise $\mathbf{1}$ and r are linearly independent. On $\text{span}\{\mathbf{1}, r\}$, using $\mathbf{1}^\top \mathbf{1} = K$, $\mathbf{1}^\top r = 1$, and $r^\top r = \|r\|_2^2 =: s$, the operator M has matrix representation $\begin{pmatrix} -1 & -s \\ 0 & Ks - 1 \end{pmatrix}$ in the basis $\{\mathbf{1}, r\}$, with eigenvalues -1 and $Ks - 1$. Hence the eigenvalues of $J^\top J$ are $\{1 \text{ (multiplicity } K - 2), 0, Ks\}$ away from the uniform point, with the same top eigenvalue formula at the uniform point. Thus

$$\|J\|_2^2 = \lambda_{\max}(J^\top J) = \max(1, Ks) = K\|r\|_2^2,$$

since $Ks \geq 1$ for a probability vector (as $\|r\|_2^2 \geq 1/K$). Because $\|r\|_2^2 \leq \|r\|_1 = 1$ on the simplex, with equality only at a vertex,

$$\|J\|_2 = \sqrt{K} \|r\|_2 \leq \sqrt{K},$$

attained exactly when r is one-hot and equal to 1 when $r = \mathbf{1}/K$. We keep this factor explicit rather than absorbing it into an unnamed constant. Combining,

$$\|J^\top W_2^\top (\text{diag}(p) - pp^\top) W_2 J\|_2 \leq \|J\|_2^2 \cdot \frac{1}{2} \sigma_{\max}(W_2)^2 \leq \frac{K}{2} \sigma_{\max}(W_2)^2.$$

□

The leading factor is $\sigma_{\max}(W_2)^2$, modulated by an NLS-Jacobian factor bounded by $K/2$ that depends only on the number of components. Two features distinguish this from Lemma 1: the bound scales with the learned spectrum of the corridor W_2 , and it is therefore parameter-dependent and time-varying rather than a fixed data constant. The NLS factor is the parameter-independent cost of calibration; the corridor factor is the source of the parameter dependence. This bounds the Gauss–Newton term only. The NLS second-order term $\sum_l (\partial L_{\text{CE}} / \partial y_l) \nabla_d^2 y_l$ need not be small because it carries the incoming head gradient, and the empirical measurements in Section 8 include the full composed curvature. The claim is not that the full composed curvature is controlled by $\sigma_{\max}(W_2)^2$, but that composition introduces a $\sigma_{\max}(W_2)^2$ dependence that the isolated site does not have.

Proposition 3 states a loss of guarantee, not a lower bound. It shows that the parameter-independent $1/2$ bound of the local LSE site no longer controls the composed supervised corridor. It does not prove that the curvature must be large for every possible W_2 or every point in training. The empirical curvature measurements in Section 8 carry that claim for the studied networks: the composed CE-corridor curvature is larger, non-monotone, and parameter-dependent.

We state the proposition without LayerNorm. The measurements are taken on the actual model, including LayerNorm, so its effect is included in the reported curvature. A proof with LayerNorm would insert another per-sample Jacobian and additional input-dependent terms; it would not restore the parameter-independent LSE guarantee.

Scope of the claim. The propositions above establish sufficient conditions for preserving the local EM conditioning guarantee: private affine component parameters, an unmixed LSE objective, and monotone per-coordinate kernels. They also show that an intervening shared, sign-indefinite learned map does not preserve the uniform guarantee. They do not prove necessity for arbitrary architectures and do not provide a convergence theorem for the composed system.

4.4 Operations That Preserve or Destroy the Invariant

The preceding propositions can be summarized by how operations treat the simplex-valued responsibility gradient.

Preserving operations. Private per-component parameters preserve the alignment between responsibility r_j and parameter block θ_j . Per-coordinate monotone maps preserve sign and component identity, changing only local scales. Smooth bounded kernels preserve parameter-independent curvature up to constants. Stochastic maps are a partial first-order preservation class: if A is row-stochastic, then pulling a nonnegative unit-mass vector back through $A^\top$ preserves nonnegativity and total mass because $\mathbf{1}^\top A^\top g = (A\mathbf{1})^\top g = \mathbf{1}^\top g$.

Destroying operations. Dense sign-indefinite learned maps do not preserve nonnegativity, normalization, or component alignment. Objective mixing replaces a single posterior gradient with a sum of gradients from different objectives; the sum is generally not the posterior of any one latent-variable model. Hard gates such as ReLU can turn small responsibility into exactly zero gradient, creating absorbing states. Shared lower parameters couple the component updates and destroy the separated M-step structure.

4.5 Classical EM Connection

The last point is the link to classical EM. This is a structural parallel, not a claim that supervised cross-entropy is an expected complete-data objective. In the separable case, the E-step produces responsibilities and the expected complete-data objective separates:

$$Q(\theta) = \sum_{j=1}^K Q_j(\theta_j).$$

Each component owns its own parameters, and the M-step decouples into independent per-component optimizations. This separation is the source of EM’s algorithmic structure.

If all components depend on a shared lower map, the separation disappears. Writing the lower parameters as ψ , the complete-data terms have the form $Q_j(\theta_j, \psi)$, so

$$Q(\theta, \psi) = \sum_j Q_j(\theta_j, \psi)$$

is no longer separable in ψ . The M-step becomes a single coupled optimization problem. Backpropagation through shared lower layers is the neural version of this classical shared-parameter failure, and the conjugated Hessian in Proposition 3 is its expression in coordinate space.

This structural account must be reconciled with the graded empirical result. The claim is not that sharing alone always destroys the effective conditioning of the full update. Rather, sharing

Classical EM failure	Mechanism of failure	Neural counterpart
Shared component parameters	Q stops separating; the M-step is coupled	Backpropagation through shared lower maps
Non-separable / generalized EM	No closed-form independent M-step	Component energies computed by deep maps
Singular components	Variance collapses and likelihood becomes unbounded	LSE-only collapse without volume control
Hard assignment	Zero-responsibility components cannot recover	Dead ReLU units
High missing information	Diffuse responsibilities slow convergence	Low 1/2-saturation and weak competition

Table 1: Classical EM failure modes and their neural counterparts.

creates a foreign CE-corridor contribution whose curvature is no longer controlled by the local LSE bound. The update at W_1 contains both this contribution and the local EM contribution:

$$\nabla_{W_1} L_{\text{total}} = \nabla_{W_1} L_{\text{CE}} + \lambda \nabla_{W_1} L_{\text{aux}}.$$

When the CE contribution dominates, the effective landscape is the non-separable shared-parameter landscape. When the EM contribution dominates, the same graph can behave like the isolated EM site. The structural result explains why the foreign term is ill-conditioned; the dominance result explains when that term matters.

5 Experimental Setup

The experiments use deliberately small supervised networks. The goal is not to improve MNIST accuracy but to make the location and failure mode of implicit EM measurable. The architecture is fixed across experiments except for the controlled ablations described below.

5.1 Model Family

The base model is a two-layer MNIST classifier with an intermediate component site:

$$x \in \mathbb{R}^{784} \rightarrow \text{Linear}(784, K) \rightarrow \text{ReLU} \rightarrow d \rightarrow \mathcal{A}(d) \rightarrow \text{Linear}(K, 10) \rightarrow \text{LayerNorm} \rightarrow L_{\text{CE}}.$$

Here K is the number of intermediate components, d is the vector of intermediate energies, and $\mathcal{A}$ is the ablated intermediate transform. Depending on the experiment, $\mathcal{A}$ is either the identity map or NegLogSoftmin:

$$\text{NLS}(d)_j = d_j + \log \sum_k \exp(-d_k).$$

When NLS is present, the site has the LSE partition function and the competitive Jacobian needed for a local implicit-EM interpretation. When NLS is absent, the same downstream classifier receives the raw ReLU-produced coordinates. Experiment 1 includes a volume-control arm without NLS to separate the effect of the calibrated NegLogSoftmin activation format from generic variance and decorrelation regularization.

The second linear map, W_2 , is the learned dense corridor from intermediate components to class logits. It is the map whose pullback removes the uniform conditioning guarantee in Proposition 3.

Because d is produced by a ReLU after the first affine map, the trained model falls under the smooth/ReLU kernel caveat following Proposition 2, not the private-affine case alone. The dead-unit metric below is therefore also a ReLU absorbing-state diagnostic. LayerNorm is kept because it stabilizes the logits seen by CE in the supervised baseline; the theory omits it, and the curvature measurements include its effect.

5.2 Local Auxiliary Objective

The local EM objective is applied to the intermediate distances d . In the causal experiment it is

$$L_{\text{aux}}(d) = L_{\text{LSE}}(d) + L_{\text{var}}(d) + L_{\text{tc}}(d).$$

The variance term L_{var} is the anti-collapse term: it penalizes components whose activations have too little variance over the batch. The decorrelation term L_{tc} is the anti-redundancy term: it penalizes correlated component responses. Together they implement the volume-control mechanism tested in the decoder-free encoder study and are related to the variance and redundancy-reduction terms used in VICReg and Barlow Twins [4, 35].

The supervised objective is

$$L_{\text{total}} = L_{\text{CE}} + \lambda L_{\text{aux}},$$

with $\lambda = 0.001$ in the initial ablation and capacity experiments unless otherwise stated. In Experiments 1–3, the auxiliary loss uses the NLS and volume-control terms specified by each ablation. In Experiment 4, the LSE term is included explicitly so that the stop-gradient arm trains W_1 by the same local EM objective as the single-layer model.

5.3 Experiment 1: Volume-Control Ablation

Experiment 1 tests which pieces of the intermediate EM site are needed. The architecture uses $K = 25$ components, MNIST, 50 training epochs, Adam with learning rate 10^{-3} , $\lambda = 0.001$, and 10 seeds (42, ..., 51). Seeds control random initialization, minibatch order, and any probe-training randomness in experiments that use probes. Metrics are averaged over the final 10 epochs and evaluated on the test set unless otherwise stated. The five configurations are:

Configuration	NLS	L_{var}	L_{tc}
Baseline	—	—	—
NLS only	✓	—	—
NLS + Var	✓	✓	—
NLS + Var + Decorr	✓	✓	✓
Var + Decorr only	—	✓	✓

This ablation separates three questions: whether supervision alone prevents intermediate collapse, whether competition without volume control suffices, and whether volume control acts as generic regularization or specifically stabilizes an EM site.

5.4 Experiment 2: Capacity Sweep

Experiment 2 tests whether the at-site repair persists across width. It compares the baseline against NLS plus full volume control at

$$K \in \{16, 25, 36, 49, 64\}.$$

Each model is trained on MNIST for 50 epochs with Adam at learning rate 10^{-3} , $\lambda = 0.001$, and 5 seeds (42, ..., 46). Metrics are again evaluated on the test set and averaged over the final 10 epochs. This experiment distinguishes structural health from benchmark accuracy: volume control can eliminate dead units and reduce redundancy even when the fixed regularization weight is not accuracy-optimal at larger widths.

5.5 Experiment 3: Optimizer Sweep

Experiment 3 asks whether the single-site optimization signature survives ordinary supervised composition. It fixes the NLS + full-volume-control model with $K = 25$ and $\lambda = 0.001$, and sweeps optimizer and learning rate:

$$\text{optimizer} \in \{\text{SGD}, \text{Adam}\}, \quad \eta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}.$$

Each setting is trained for 50 epochs on MNIST with 3 seeds (42, 43, 44). Reported metrics are evaluated on the test set and averaged over the final 10 epochs. The comparison is aligned with the single-layer implicit-EM signature: learning-rate sensitivity, Adam advantage, and whether lower auxiliary loss corresponds to better features.

5.6 Experiment 4: Mechanism and Causal Test

Experiment 4 measures the mechanism behind the optimizer result. The model is

$$x \rightarrow \text{Linear}(784, 25) \rightarrow \text{ReLU} \rightarrow d \rightarrow \text{NLS}(d) \rightarrow \text{Linear}(25, 10) \rightarrow \text{LayerNorm} \rightarrow L_{\text{CE}},$$

with

$$L_{\text{aux}}(d) = L_{\text{LSE}}(d) + L_{\text{var}}(d) + L_{\text{tc}}(d).$$

Four arms differ only in how the supervised and local objectives reach W_1 :

Arm	Total loss	CE reaches W_1 ?
$\lambda = 0.001$ joint	$L_{\text{CE}} + 0.001L_{\text{aux}}$	✓
$\lambda = 0.03$ joint	$L_{\text{CE}} + 0.03L_{\text{aux}}$	✓
$\lambda = 1$ joint	$L_{\text{CE}} + L_{\text{aux}}$	✓
stop-gradient	$L_{\text{CE}}(\text{head}(y.\text{detach}())) + L_{\text{aux}}$	—

The stop-gradient arm keeps the same data, labels, architecture, and supervised head training, but prevents the CE gradient from reaching W_1 . This makes it the causal control for the corridor.

For the conditioning sweep, each arm is trained with SGD at

$$\eta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$$

for 40 epochs and 3 seeds. Feature quality is measured after training by a fixed-protocol linear probe trained on the frozen intermediate distances d . The probe is a single linear classifier trained with Adam at learning rate 10^{-3} for 15 epochs, batch size 256, with its own fixed random seed. This decouples the measured representation from how well the supervised head trained at a particular learning rate [3].

For gradient competition, the training loop measures

$$\|\nabla_d L_{\text{CE}}\|, \quad \lambda \|\nabla_d L_{\text{aux}}\|, \quad \cos(\nabla_d L_{\text{CE}}, \nabla_d L_{\text{aux}}).$$

These measurements are taken during Adam training at learning rate 10^{-3} , while the conditioning sweep uses SGD. We therefore report these ratios as a site-level gradient diagnostic rather than as a matched-optimizer measurement; the matched-SGD version is not reported here.

For curvature tracking, power iteration with exact Hessian-vector products estimates the spectral norm of the Hessian with respect to d for both the LSE term and the CE corridor through NLS and the head. Measurements use a fixed 512-example batch every 2 epochs, and $\sigma_{\max}(W_2)$ is recorded alongside the curvature estimates [25].

For the private-affine scale check in Proposition 1, we also compute the data constant $\mathbb{E}\|\tilde{x}\|^2$ using the raw $[0, 1]$ MNIST pixels with a bias coordinate. Section 7 compares the implied LSE descent scale to the stop-gradient learning-rate sweep.

5.7 Metrics

Representation health is measured by four quantities. A dead unit is a component whose activation variance is below 0.01 over the evaluation batch. Minimum variance records the weakest component variance. Redundancy is $\|\text{Corr}(d) - I\|_{F,\text{off}}^2$, the squared Frobenius norm of the off-diagonal correlation matrix among component responses. Responsibility entropy is the mean sample entropy

$$\frac{1}{N} \sum_{i=1}^N \left[- \sum_{j=1}^K r_{ij} \log(r_{ij} + 10^{-8}) \right], \quad r_i = \text{softmax}(-d_i),$$

and measures how sharp the local component assignments are.

Optimization conditioning is measured by learning-rate spread, convergence-time stability, Adam versus SGD gaps, gradient dominance ratios, cosine similarity between CE and auxiliary gradients, and Hessian spectral norms with respect to the intermediate distances. For cosine measurements, we compare the observed values to the random-vector scale $1/\sqrt{K}$, not to zero alone.

For convergence time, the reported LSE-stabilization metric is interpreted as an EM-site stability measure; in CE-dominated arms the classifier may train well even if the local LSE site does not stabilize.

5.8 Experimental Logic

The experiments follow the paper’s taxonomy. Experiments 1–2 test at-site failures: collapse, dead units, and redundancy. Their question is whether local volume control repairs these failures inside a supervised network. Experiment 3 tests whether the single-site conditioning signature survives ordinary supervised composition. Experiment 4 separates structure from dominance: the dense corridor creates an ill-conditioned CE-corridor contribution, while the measured CE/EM gradient ratio determines whether that contribution controls the effective landscape.

6 Results I: At-Site Failures Are Repaired Locally (Experiments 1–2)

The theory separates two kinds of failure. Structural failures arise when a responsibility gradient is pulled through a shared learned corridor. At-site failures occur at the EM interface itself: components die, collapse, or become redundant. The first two experiments test the at-site half. They ask whether supervision supplies enough intermediate volume control, and whether the variance and decorrelation terms repair the same failures inside a supervised network that they repair in the single-layer model.

6.1 Supervision Does Not Supply Intermediate Volume Control

Experiment 1 ablates the intermediate EM site at $K = 25$. All models share the same supervised classifier after the intermediate layer; only NLS and the local volume-control terms are changed. Table 2 reports the summary over the final 10 epochs and 10 seeds. The structural metrics are

Config	Dead Units	Min Var	Redundancy	Resp. Entropy	Accuracy
Baseline	1.40 ± 0.49	0.000 ± 0.000	20.5 ± 2.8	3.08 ± 0.02	96.09 ± 0.07
NLS only	1.20 ± 0.98	0.264 ± 0.408	24.9 ± 3.0	2.93 ± 0.02	96.14 ± 0.17
NLS + Var	0.00 ± 0.00	11.74 ± 3.08	187.0 ± 38.7	2.09 ± 0.09	95.75 ± 0.21
NLS + Var + Decorr	0.00 ± 0.00	7.84 ± 1.17	13.9 ± 0.5	2.46 ± 0.04	96.34 ± 0.11
Var + Decorr only	0.10 ± 0.30	3.55 ± 1.49	13.3 ± 0.9	2.65 ± 0.07	96.40 ± 0.10

Table 2: Volume-control ablation at $K = 25$. Metrics are defined in Section 5; accuracy is in percent. All values are mean $\pm$ standard deviation over seeds.

intentionally close to the auxiliary objectives: minimum variance tracks the anti-collapse term, and redundancy tracks the decorrelation term. We therefore read them as targeted repair diagnostics, not as independent measures of representation quality. The cross-arm pattern is the important check: variance-only eliminates dead units while worsening redundancy, and the full objective repairs both failures together.

The baseline has minimum variance zero across all seeds and an average of 1.4 dead units out of 25. Supervised cross-entropy trains the output layer, but the gradient reaching the hidden layer through W_2 does not prevent intermediate collapse. This is the first at-site result: labels provide structure at the output interface, but not volume control for an intermediate component site.

6.2 Competition Without Volume Control Is Insufficient

NLS alone introduces the LSE partition function and the competitive Jacobian at the intermediate site. If competition by itself stabilized the representation, NLS-only should improve the health metrics. It does not. Accuracy is unchanged relative to the baseline, dead units remain, and redundancy increases from 20.5 to 24.9. The local EM geometry is present, but without volume control it has no mechanism for keeping every component alive and distinct.

This bears directly on the paper’s central distinction. The failure here is not that the EM identity is absent; the NLS site supplies the identity. The failure is that an EM site without volume control has the same degeneracies as an unregularized mixture model.

6.3 Partial Volume Control Is Worse Than None

The variance-only arm isolates the anti-collapse half of volume control. It works on the metric it directly targets: dead units fall to zero and minimum variance rises to 11.74. But it is the worst configuration overall. Redundancy rises to 187.0, roughly nine times the baseline, and accuracy falls to 95.75%.

The mechanism is straightforward. A variance penalty forces every component to move, but without decorrelation the cheapest way for all components to have high variance is for them to move together. The representation becomes alive but redundant. The low responsibility entropy in this arm, 2.09, should not be read as healthy specialization: it is sharp competition among nearly duplicated components.

This replicates the decoder-free encoder ablation inside a supervised network [24]. The diagonal and off-diagonal parts of volume control are not interchangeable. Anti-collapse without anti-redundancy can be worse than neither.

6.4 Full Volume Control Repairs the Site

Adding decorrelation fixes the variance-only pathology. The full NLS + Var + Decorr model has zero dead units, high minimum variance, and the lowest redundancy among the EM-site configurations. Redundancy falls to 13.9, and accuracy rises to 96.34% with a tight seed standard deviation.

The accuracy improvement is not the main claim. The main claim is that local volume control repairs local structural health. The full model keeps every component alive and distinct while preserving the calibrated responsibility structure introduced by NLS.

6.5 Volume Control Without NLS Is Generic Regularization, Not the Same Mechanism

The Var + Decorr only arm is an important control. It matches the full EM-site model on accuracy and redundancy, showing that volume control is also a useful regularizer. But the mechanism is formally different by construction: without NLS, the model has no calibrated distance-to-responsibility interface at the hidden site. The measured geometry also differs. Minimum variance is lower (3.55 versus 7.84), and the entropy of $\text{softmax}(-d)$ is higher (2.65 versus 2.46). In the no-NLS arm this entropy is only a descriptive pseudo-responsibility statistic, not evidence that the raw ReLU coordinates were trained as calibrated responsibilities.

This control constrains the interpretation. The paper does not claim that every gain is EM-specific; some gains are ordinary regularization. The EM-specific claim is narrower: when a local EM site exists, the same volume-control terms that stabilize mixture components stabilize the site’s responsibilities and component geometry.

6.6 The Capacity Sweep: Structural Health Persists Across Width

Experiment 2 tests whether the ablation result is specific to $K = 25$. It compares the baseline against NLS plus full volume control for $K \in \{16, 25, 36, 49, 64\}$, using 5 seeds per width. Table 3 summarizes the key structural and accuracy metrics. The dead-unit result is width-invariant. Every baseline width has dead units, and the dead fraction is stable at roughly 5–7%. Full volume control eliminates dead units at every width. This is not a small-architecture artifact: the baseline routes around dead capacity, while volume control keeps all components active.

Redundancy also scales in the predicted direction. In the baseline, redundancy grows roughly linearly with width, from 11.0 at $K = 16$ to 69.0 at $K = 64$. Full volume control lowers redundancy at every width and sharply reduces the seed-to-seed variance. At $K = 64$, the redundancy standard deviation falls from 7.2 to 0.8.

6.7 Accuracy Is Secondary and Capacity-Dependent

The accuracy pattern is more nuanced. With the fixed λ used in the sweep, volume control improves accuracy at $K = 16$ and $K = 25$, is neutral around $K = 36$, and slightly hurts at $K = 49$ and $K = 64$. This is not a failure of the structural claim. The fixed regularization weight is not calibrated per width, and one plausible interpretation is that the larger models can use some correlated extra units as supervised boundary refiners. Under that interpretation, a decorrelation penalty treats useful cooperation as redundancy, even when it helps the task.

K	Config	Dead	Dead Frac.	Min Var	Redundancy	Accuracy
16	Baseline	1.0 ± 0.9	6.3%	0.097 ± 0.119	11.0 ± 2.2	95.19 ± 0.19
16	Full VC	0.0 ± 0.0	0.0%	5.22 ± 1.09	9.1 ± 0.9	95.32 ± 0.21
25	Baseline	1.4 ± 0.5	5.6%	0.000 ± 0.000	19.0 ± 1.3	96.07 ± 0.07
25	Full VC	0.0 ± 0.0	0.0%	7.76 ± 1.21	13.8 ± 0.5	96.35 ± 0.12
36	Baseline	2.4 ± 1.9	6.7%	0.030 ± 0.059	31.7 ± 5.1	96.63 ± 0.14
36	Full VC	0.0 ± 0.0	0.0%	9.51 ± 0.10	21.1 ± 1.5	96.60 ± 0.10
49	Baseline	2.5 ± 0.9	5.0%	0.000 ± 0.000	48.1 ± 3.6	97.04 ± 0.09
49	Full VC	0.0 ± 0.0	0.0%	12.65 ± 1.55	29.7 ± 0.5	96.87 ± 0.16
64	Baseline	4.1 ± 1.3	6.3%	0.000 ± 0.000	69.0 ± 7.2	97.29 ± 0.06
64	Full VC	0.0 ± 0.0	0.0%	15.33 ± 0.92	41.2 ± 0.8	97.00 ± 0.09

Table 3: Capacity sweep. Accuracy is reported in percent. Full VC denotes NLS plus variance and decorrelation penalties. These are separate 5-seed capacity runs, so the $K = 25$ rows need not exactly match the 10-seed $K = 25$ ablation in Table 2.

The result should therefore be read as two regimes. In the capacity-constrained regime, dead and redundant units are expensive; volume control improves both structure and accuracy. In the capacity-abundant regime, overparameterization can absorb dead units, and correlated features may help supervised boundary refinement. Volume control still produces healthier component geometry, but the task accuracy trade-off depends on λ and width.

6.8 Conclusion of Results I

Experiments 1–2 establish the at-site half of the paper. The supervised gradient does not supply volume control to intermediate components. Competition without volume control is insufficient. Partial volume control can be pathological. Full volume control repairs collapse and redundancy locally, and the repair persists across width.

This is the part of implicit EM that transfers through supervised networks: local mixture degeneracies can be fixed by local volume control. The next results ask whether the optimizer conditioning transfers as well.

7 Results II: Conditioning Does Not Survive Composition (Experiment 3)

Results I showed that volume control transfers to an intermediate supervised site: it eliminates dead units and reduces redundancy even though the network is trained for classification. Experiment 3 asks whether the optimization conditioning from the single-layer implicit-EM model transfers as well. The answer is no. Once the local site is embedded behind an ordinary supervised head, SGD recovers its learning-rate sensitivity and the advantage of adaptive methods reappears.

This experiment uses the full EM-site model from Results I: NLS plus variance and decorrelation, with $K = 25$ and $\lambda = 0.001$. It is therefore not a plain baseline optimizer sweep; the question is whether the full local EM site preserves the single-site optimizer signature after supervised composition. We train with SGD and Adam over learning rates $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ for 50 epochs on MNIST. Table 4 reports metrics averaged over epochs 41–50 and 3 seeds.

Opt.	LR	Acc.	CE	Reg.	Min Var	Redundancy
SGD	10^{-4}	83.83 ± 4.96	0.868	107.5	0.015 ± 0.007	27.5 ± 0.9
SGD	10^{-3}	93.69 ± 0.26	0.275	69.9	0.065 ± 0.013	19.6 ± 1.3
SGD	10^{-2}	95.63 ± 0.12	0.155	19.2	0.436 ± 0.031	17.7 ± 0.5
SGD	10^{-1}	96.23 ± 0.09	0.133	-34.3	4.01 ± 0.34	16.2 ± 1.3
Adam	10^{-4}	95.34 ± 0.33	0.170	24.6	0.324 ± 0.108	18.0 ± 0.5
Adam	10^{-3}	96.38 ± 0.11	0.133	-57.3	7.80 ± 1.55	13.8 ± 0.6
Adam	10^{-2}	96.37 ± 0.12	0.160	-158.3	423.2 ± 40.6	11.9 ± 0.5
Adam	10^{-1}	96.58 ± 0.05	0.179	-265.9	30628.4 ± 2967.6	11.9 ± 0.4

Table 4: Optimizer sweep for the full EM-site model. All metrics are evaluated on the test set and averaged over the final 10 epochs and 3 seeds; accuracy is in percent. Reg. is the unweighted local volume-control objective $L_{\text{var}} + L_{\text{tc}}$ before multiplication by λ . It can be negative because $L_{\text{var}} = -\sum_j \log(\text{Var}(d_j) + \epsilon)$ is negative when component variances exceed one.

7.1 The Single-Site Optimizer Signature Disappears

The single-layer implicit-EM model had an unusual optimization signature: SGD was insensitive to learning rate over a wide range, Adam offered little advantage, and convergence time was largely fixed [24]. Those properties do not survive ordinary supervised composition.

With SGD, accuracy spans 83.83% to 96.23% across the four learning rates, a 12.4 point range. At 10^{-4} , the model has not converged by epoch 50; at 10^{-3} , it reaches 93.69% but remains well below the best settings. The learning-rate axis is therefore active again. In this grid, the evidence is final-accuracy and convergence-speed sensitivity, not an observed instability edge at high learning rates: SGD improves monotonically through 10^{-1} . The local EM site and its volume control are present, but they do not make the composed supervised landscape learning-rate insensitive over the tested range.

This is the empirical counterpart of Proposition 3. Composition does not prove that curvature must be large at every point, but it removes the parameter-independent guarantee that explained the single-site behavior. Experiment 3 shows that, in this supervised network, the missing guarantee matters.

7.2 The Adaptive-Optimizer Advantage Returns

Adam improves the low-learning-rate regime. At learning rate 10^{-4} , Adam reaches 95.34%, close to SGD at 10^{-2} and far above SGD at the matched learning rate. At 10^{-3} , Adam reaches 96.38%, matching or slightly exceeding SGD at 10^{-1} . The CE loss shows the same pattern: Adam finds low supervised loss efficiently at learning rates where SGD is still undertrained.

This does not contradict Results I. Volume control is still repairing the intermediate site. The point is narrower: the optimizer no longer sees the private, responsibility-weighted landscape of the single-layer EM model. The gradient reaching W_1 is the sum of a local auxiliary contribution and a supervised contribution pulled back through the learned head. That composed gradient behaves like an ordinary deep-network gradient, and adaptive scaling is useful.

7.3 The Local Site Still Responds to Volume Control

Although the conditioning signature disappears, the local volume-control objective is not inert. As learning rate increases, the structural metrics move in the expected direction. For SGD, minimum

variance rises from 0.015 to 4.01 and redundancy falls from 27.5 to 16.2. For Adam, redundancy falls to 13.8 at 10^{-3} and to 11.9 at the higher learning rates. The auxiliary objective is shaping the representation.

The distinction is therefore between structural repair and optimizer conditioning. Volume control can repair at-site failures while failing to control the global training dynamics. This is the central split of the paper: collapse prevention transfers locally, but EM conditioning does not automatically propagate through the supervised corridor.

7.4 Loss-Feature Decoupling Partially Replicates

The high-learning-rate Adam runs recover one part of the single-layer phenomenology: lower auxiliary loss need not mean better task features. From Adam 10^{-3} to 10^{-1} , the regularization loss decreases from -57.3 to -265.9 and minimum variance grows from 7.80 to 30628.4. Accuracy, however, barely moves: 96.38%, 96.37%, and 96.58%.

This is not a useful improvement in the representation. It shows that the volume-control objective has escape directions that are largely orthogonal to classification accuracy once the useful level of spread and decorrelation has been reached. The negative regularization values are not a sign error: the variance term is $-\sum_j \log(\text{Var}(d_j) + \epsilon)$, so large component variance can drive it downward. LayerNorm after the head may make this scale growth easier for CE to tolerate, but the sweep does not isolate that mechanism. Adam at 10^{-3} is therefore the operating point used below: it sits below the runaway regime while retaining high accuracy, low redundancy, and finite variance.

7.5 What Experiment 3 Does and Does Not Establish

Experiment 3 establishes the observational boundary. The same local objective that repairs collapse in Results I does not preserve the single-site optimizer signature under

8 Results III: The Mechanism and Causal Test (Experiment 4)

Experiment 3 established the boundary observationally; Experiment 4 asks why. It tests three mechanism predictions—CE-corridor dominance at small λ , the curvature dichotomy, and causal restoration under stop-gradient—summarized against outcomes in the ledger below.

8.1 Predictions Ledger

Table 5 summarizes the predictions tested in the supervised study and the strengthening checks still open. Because the claims are causal, the ledger separates predictions, measurements, status, and the one interpretation corrected after measurement.

8.2 Experimental Arms

The causal experiment uses the model and auxiliary objective from Section 5 with $K = 25$. Four arms differ only in how the supervised and local objectives reach W_1 : The conditioning sweep trains each arm with SGD over a $1000\times$ learning-rate range, $\eta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, using 3 seeds. Feature quality is measured by a fixed-protocol linear probe trained on frozen intermediate distances. This separates representation quality from the training success of the supervised head at a particular learning rate.

8.3 Cutting the Corridor Restores Learning-Rate Insensitivity

The stop-gradient arm restores the most visible optimizer signature from the single-layer implicit-EM study [24]: feature quality becomes nearly insensitive to learning rate. Table 7 shows linear-probe accuracy across the $1000\times$ SGD sweep. With ordinary joint training at $\lambda = 0.001$, probe accuracy varies by 10.9 points across the learning-rate range. When the CE gradient is prevented from reaching W_1 , the spread falls to 1.6 points. The network is still supervised: the same labels, CE loss, architecture, and head training remain. Only the corridor gradient is removed. Supervision itself therefore does not destroy the conditioning signature; the pullback through the corridor does. As in Experiment 3, this spread is a convergence-speed and final-probe-accuracy diagnostic over the tested grid, not evidence that any arm has reached an instability edge at high learning rates.

The $\lambda = 1$ joint arm is equally important. It is fully connected: the CE gradient still reaches W_1 . Yet it is indistinguishable from stop-gradient on the conditioning metric. Connectivity is therefore not the decisive variable. The corridor creates a foreign, non-EM contribution, but that contribution controls the effective landscape only when it dominates the local EM contribution. The same point appears in the feature quality: even with label gradients still flowing, the $\lambda = 1$ arm probes at the stop-gradient level, not at the CE-dominated level.

The private-affine theory also gives a coarse parameter-space scale check. With the MNIST preprocessing used here, raw pixels lie in $[0, 1]$ and the augmented input satisfies $\mathbb{E}\|\tilde{x}\|^2 = 88.8$ on the training set. The LSE bound from Proposition 1 is therefore $L \leq \frac{1}{2}\mathbb{E}\|\tilde{x}\|^2 = 44.4$, corresponding to a descent-lemma scale $2/L \approx 0.045$. The stop-gradient arm remains stable through $\eta = 0.1$, which is the same order of magnitude. This should be read only as a sanity check: the experiment uses ReLU distances and the full L_{aux} objective, so it is not a formal test of the LSE-only private-affine bound.

The intermediate value $\lambda = 0.03$ lies between the two regimes. Its spread is 5.8 points, roughly halfway between the CE-dominated and EM-dominated arms. This is the first evidence for the graded claim: the transition is not a phase change at the presence or absence of a graph edge. It follows dominance.

8.4 The EM Site Stabilizes Only When the EM Path Is Live

We also track the epoch at which test L_{LSE} reaches 95% of its final value. This is a stability metric for the local EM site, not a general classifier-training metric. In the CE-dominated arm, the classifier can train successfully while the local LSE objective receives little effective training of its own. Under stop-gradient, the local site stabilizes in a narrow band, roughly 22–32 epochs across a $1000\times$ learning-rate range. Under CE-dominated joint training, the same metric is erratic and sometimes reaches the training budget. This supports the same conclusion as the probe sweep: when W_1 is trained by the local EM objective, the site has the fixed-time behavior of the single-layer model; when the CE corridor dominates, the local EM site does not stabilize as an autonomous mixture. Because the metric is self-normalized, the safe interpretation is the contrast in stability across learning rates, not a claim about a shared absolute LSE threshold.

8.5 Gradient Dominance, Not Gradient Opposition

The mechanism is visible directly at the distance interface. In the reported diagnostic, the CE and auxiliary gradients with respect to d are measured on the same batch during Adam training at learning rate 10^{-3} . The conditioning sweep uses SGD, so the table should be read as a site-level

gradient diagnostic rather than a matched-optimizer measurement. Table 9 reports the ratio

$$\frac{\|\nabla_d L_{\text{CE}}\|}{\lambda \|\nabla_d L_{\text{aux}}\|}$$

and the cosine between the two gradient directions. This measurement corrects an earlier heuristic interpretation. Since $\lambda = 0.001$, one might expect a $1000\times$ ratio. The measured ratio is $67\text{--}95\times$ late in training, because the raw auxiliary gradient is larger than the raw CE gradient. The full trajectory is wider, with early transients before the CE corridor takes over. The conclusion is unchanged: CE dominates at the operating point, but the causal variable is the measured gradient ratio, not λ itself.

The cosine measurement shows no detectable alignment between the two paths. In $K = 25$ dimensions, random directions have cosine standard deviation of approximately $1/\sqrt{25} = 0.2$, so a cosine near zero should not be interpreted as special high-dimensional orthogonality. The conservative statement suffices: the EM and CE gradients are not detectably aligned or opposed. The EM signal is diluted by a larger foreign signal, not cancelled by an antagonistic one. The observed ranges are smaller than this coarse random-direction scale; batch averaging and the shared NLS interface likely change the relevant null, so we do not interpret that undershoot further.

The prediction is that the dominance ratio is a property of the loss geometry at the site, not an Adam-specific artifact. That remains open until the same CE/EM ratio is measured under the matched SGD sweep used for the conditioning result. The current Adam measurement supports the mechanism, but it does not close the matched-optimizer question.

8.6 Curvature: The LSE Bound Holds, the CE Corridor Is Large

Proposition 3 says that composition removes the uniform LSE guarantee; it does not prove that the composed curvature is necessarily large. The curvature measurements answer that empirical question. Power iteration on exact Hessian-vector products estimates the spectral norm of the Hessian with respect to the intermediate distances d for both the local LSE term and the CE corridor through NLS and the head [25].

The local LSE curvature obeys the theory at every measured point: it stays below $1/2$ in every arm and seed. In the stop-gradient arm, it converges to $0.4992\text{--}0.4997$, essentially saturating the bound. The bound is achieved when responsibility mass concentrates equally on two components. Under protected local-EM training, the mixture sharpens until cluster boundaries become two-way ties, and the Hessian detects exactly this geometry.

By contrast, the CE-corridor curvature in a representative joint $\lambda = 0.001$ trajectory takes values including 6.5 , 17.2 , and 7.8 across training. At the same coordinates, this is roughly $13\text{--}100\times$ larger than the LSE curvature, non-monotone, and consistent with dependence on the learned spectrum of W_2 . This is the empirical half of the theory: composition removes the parameter-independent guarantee, and in the trained network the resulting curvature is in fact large and changing.

The $1/2$ bound is therefore more than a worst-case constant. Distance from $1/2$ acts as a site-level diagnostic of mixture sharpness. Under EM-dominated training it saturates; under CE-dominated training, the LSE curvature stagnates well below the bound, ending between 0.158 and 0.236 across seeds. This indicates diffuse responsibilities and an under-formed local mixture.

8.7 Conditioning Is Restored at a Cost

The EM-dominated arms are better conditioned but less task-aligned. Their frozen features probe at about 88% , while the CE-dominated arm reaches about 96% . This is expected in the stop-

gradient arm: W_1 does not receive label information. The point is not that unsupervised local features beat supervised features; they do not.

The point is that conditioning is a real property with a real cost. The EM regime provides learning-rate robustness, seed-to-seed reproducibility, stable local convergence, and zero dead units; it does not provide output accuracy. This is why a denser λ -resolution curve is the natural strengthening analysis: the trade-off should ultimately be shown as a continuum along the measured CE/EM gradient ratio, not as a binary comparison between supervised and unsupervised training.

9 Discussion

The results draw a boundary around the implicit-EM interpretation of neural training. The gradient-responsibility identity is real, and it has observable consequences inside a supervised network. But those consequences are local. A network can contain an EM site, benefit from local volume control at that site, and still fail to inherit the optimizer conditioning of an isolated EM model.

9.1 Local, Graded, and Structural

The three adjectives in the main claim refer to different parts of the evidence.

Implicit EM is *local*: the responsibility structure exists at an LSE or softmax site and extends to parameters only when the path preserves component identity—it does not automatically penetrate a learned dense map. The theory is the formal version of this. When the responsibility vector updates private component parameters directly, Proposition 1 gives a parameter-independent curvature bound; when it is pulled through a learned dense map, Proposition 3 shows the same bound no longer controls the composed corridor. Sections 7 and 8 show the empirical consequence: ordinary supervised composition restores learning-rate sensitivity and an Adam advantage.

Implicit EM is *graded*. The decisive variable is not whether the computation graph contains the CE corridor to W_1 . The $\lambda = 1$ joint arm keeps that corridor intact and nevertheless behaves like the stop-gradient arm on the conditioning metric. Conversely, the $\lambda = 0.001$ joint arm is CE-dominated and behaves like an ordinary supervised network. The reported CE/EM gradient-ratio diagnostic identifies the causal variable as dominance, not graph connectivity alone, although the matched-SGD version of that diagnostic remains an open strengthening check.

Implicit EM is *structural*. Volume control repairs failures at a local EM interface: dead units, collapsed variance, and redundancy. Dense learned composition creates a different failure: the foreign CE contribution reaching W_1 is not a private per-component responsibility update. The two failure classes require different repairs. At-site degeneracy is repaired by local volume control; structural loss of conditioning is repaired only by isolation, dominance, or compatible transport.

9.2 What Transfers

The part that transfers is the local mixture geometry. As Results I established, supervised learning does not by itself supply volume control to an intermediate site, and only full volume control (not competition alone, and not the anti-collapse term alone) eliminates dead units and redundancy across widths. The local EM site changes the structural health of the representation in the same direction observed in the decoder-free encoder study; Results I replicates that volume-control pattern inside a supervised network.

The Var+Decorr control, which also performs well, strengthens rather than weakens the interpretation. It separates ordinary regularization from the narrower EM-specific claim: volume control

is useful regularization on its own, and when attached to an NLS site it additionally stabilizes the calibrated distance-to-responsibility geometry.

9.3 What Does Not Transfer

The single-site optimizer signature does not transfer through ordinary dense supervised composition. In the isolated implicit-EM model, SGD was unusually learning-rate insensitive and Adam had little to correct. In the supervised network, SGD spans a wide accuracy range across learning rates and adaptive scaling becomes useful. The local objective remains present, but it no longer controls the training dynamics at W_1 when the supervised corridor dominates.

The causal experiment explains why. Cutting the CE-to- W_1 corridor restores the learning-rate-insensitive signature. Increasing λ until the EM path dominates restores the same signature without cutting the graph. The failure is therefore not supervision itself and not the existence of a second layer. It is the dominance of a composed, non-EM gradient at the parameters the local EM site was intended to condition.

9.4 Dominance Resolves the Apparent Tension

There is an apparent tension between two statements: dense backpropagation is a form of parameter sharing, yet the fully connected $\lambda = 1$ arm behaves like stop-gradient. The resolution is to separate structure from dominance.

The structural statement explains the form of the foreign contribution. Once CE is pulled back through W_2 , the gradient at W_1 is not a posterior over private hidden components, and its curvature can depend on the learned spectrum of the corridor. This is the neural version of a non-separable M-step.

The dominance statement explains whether that contribution matters. The update at W_1 is a sum:

$$\nabla_{W_1} L_{\text{total}} = \nabla_{W_1} L_{\text{CE}} + \lambda \nabla_{W_1} L_{\text{aux}}.$$

When the first term dominates, the effective landscape is the composed supervised landscape. When the second term dominates, the local EM geometry sets the behavior even though the CE corridor still exists. A denser λ -resolution curve is therefore the natural next strengthening analysis: the causal variable is the measured gradient ratio, not the nominal value of λ and not the binary presence of an edge.

The private-affine scale check points in the same direction without carrying the main claim. The MNIST data constant gives an LSE descent scale of roughly 0.045, and the stop-gradient arm is stable through learning rate 0.1. This is only an order-of-magnitude sanity check because the experiment uses ReLU distances and the full auxiliary objective, but it connects the local curvature bound to the observed robust learning-rate range.

9.5 Generative Versus Discriminative Is the Phenomenon, Not the Mechanism

One deflationary reading is that the paper rediscovers the familiar generative-versus-discriminative distinction. The experiments are compatible with that high-level framing, but they are more specific. The same architecture is trained with the same data and the same local site while the route by which CE reaches W_1 is changed. The mechanistic measurements are specific: the LSE curvature obeys the $1/2$ bound, the CE-corridor curvature is larger and changing, the reported gradient diagnostic shows CE dominance at small λ , and removing that corridor restores the conditioning signature.

The result is therefore not only that two objectives differ. It identifies the route by which the discriminative objective reaches a local EM site and the condition under which that gradient controls the update. This is the useful level of explanation for model design: it identifies which corridor must be blocked, weakened, or made compatible if local EM conditioning is to survive.

The main remaining qualifications are collected in Section 10: the matched-optimizer gradient ratio is not yet reported, the learning-rate sweeps do not find an upper instability edge, and the high-learning-rate Adam runs expose an auxiliary escape direction.

9.6 The Accuracy Trade-Off

The EM-dominated features probe around 88%, while the CE-dominated features probe around 96%. That gap should be stated plainly. Well-conditioned optimization of a less task-aligned local objective does not beat direct supervised optimization of the task objective.

This does not make the conditioning result unimportant. It clarifies what the property is for. EM conditioning provides robustness to learning-rate choice, lower seed-to-seed variability, stable local convergence, and reliable use of all components. Those benefits matter most when the local objective is not being asked to replace the supervised head at the output: for layer-local representation learning, protected intermediate objectives, or architectures where local mixture structure is itself the desired object. In an ordinary end-to-end classifier, the task head can trade away that structure for accuracy.

9.7 Pretraining and Fine-Tuning

The results suggest a concrete interpretation of discriminative fine-tuning. If a representation is first shaped by a local EM objective and then exposed to a dominant supervised corridor, fine-tuning may not simply refine the EM solution. It may move the model into a different regime whose features are more task-aligned but less locally EM-conditioned [15].

The current evidence is suggestive rather than complete. The EM-dominated and CE-dominated arms occupy different accuracy and conditioning regimes, but local gradient directions are not basin evidence. A basin or linear-mode-connectivity analysis, with hidden units aligned before interpolation, would test the stronger claim: whether the two regimes are connected by a low-loss path or separated by a barrier after accounting for permutation symmetry.

9.8 Design Implications

The practical rule is simple: create EM sites locally, and do not expect ordinary dense backpropagation to preserve their conditioning. If the goal is local mixture structure, the objective should reach the relevant parameters directly. If a supervised or foreign objective must share those parameters, its gradient must either be blocked, balanced so the EM path dominates when desired, or transported through operations that preserve the simplex structure.

This rule also clarifies the status of the two interventions. Stop-gradient is not only a training convenience in this setting; it is a structural intervention that turns a shared, non-separable update back into a local one. Conversely, increasing λ does not simply add regularization strength; it changes the effective optimization regime by changing which gradient controls W_1 .

The stochastic-map case is the natural remaining taxonomy cell. A nonnegative row-normalized corridor can preserve nonnegativity and total mass at first order, unlike an arbitrary dense signed matrix. This variant should be interpreted as a test of partial preservation, not as a headline result. If row normalization damages task accuracy enough to confound the probe, the result belongs as a caveated appendix experiment.

9.9 Relation to Attention

Companion work in preparation asks whether EM-like structure propagates through transformer depth. The locality result here gives the controlled answer that such propagation should not be assumed. Attention has known connections to associative-memory, softmax retrieval, and EM-like gradient-dynamics views [1, 28], but a standard transformer has an output CE site that owns an objective; intermediate attention softmaxes are computation inside a larger objective, not automatically owned EM sites. Their gradients are shaped by foreign downstream losses and learned linear maps.

The connection should remain narrow in this paper. The supervised study is the controlled test of the locality principle. Transformer experiments would be a less controlled test of the same design rule. The stochastic-map variant is the bridge: attention transports values through a row-stochastic map at first order, which may preserve some responsibility-like structure while still failing to preserve full optimizer conditioning through the surrounding learned projections and downstream losses.

10 Limitations and Threats to Validity

The paper makes a narrow claim about where implicit-EM conditioning is preserved and where it is lost. The evidence is intentionally controlled rather than broad. This section separates threats to the internal interpretation of the experiments from limitations on external scope.

10.1 Threats to Internal Validity

Feature quality is measured primarily by a linear probe. The main accuracy trade-off compares frozen intermediate representations with a fixed-protocol linear probe. This is the right first probe because it measures linearly accessible class information without allowing the downstream evaluator to repair the representation. It is not the only possible measure of feature quality. A kNN probe and a small MLP probe on the same frozen distances would test whether the 88% versus 96% ordering is a property of the representation or a property of the probe. If the ordering survives these checks, the interpretation strengthens: EM-dominated features are genuinely less task-aligned, not only less linearly separable.

The curvature measurements depend on numerical estimation. The curvature claims use spectral norms estimated by power iteration on fixed batches. The qualitative contrast is large: the local LSE curvature remains below $1/2$, while the composed CE-corridor curvature is much larger and non-monotone. Still, the exact reported ratios can depend on batch identity, batch size, and the number of power-iteration steps. We did not run a full sensitivity sweep over these numerical choices. The key internal check is that the LSE bound remains stable and the CE-corridor curvature remains separated from it under reasonable measurement variations.

The $1/2$ -saturation diagnostic is demonstrated at one site. The paper uses proximity to the LSE curvature bound as a diagnostic of mixture sharpness. In this study, the diagnostic behaves as predicted: EM-dominated training approaches the $1/2$ bound, while CE-dominated training leaves the local site diffuse. That supports the diagnostic locally, but not its general use across architectures. Claims about $1/2$ -saturation should therefore remain modest until the same measurement is repeated across depths, datasets, or different EM-site parameterizations.

The data-constant learning-rate scale is only a sanity check. Proposition 1 gives an LSE-only scale in terms of $\mathbb{E}\|\tilde{x}\|^2$. Results III reports this constant for MNIST and compares the implied descent-lemma scale to the stop-gradient learning-rate sweep. The agreement is only order-of-magnitude evidence. The trained model uses ReLU distances and the full auxiliary objective, not the pure private-affine LSE objective, so this comparison should not be read as a formal prediction of the largest stable learning rate.

Stop-gradient is a precise but strong intervention. The stop-gradient arm blocks the CE gradient from reaching W_1 while keeping the data, labels, architecture, CE loss, and supervised head in place. This makes it a strong causal test of the corridor-gradient hypothesis. It is also a large intervention: it changes the optimization problem for W_1 from joint supervised training to protected local training. The interpretation rests on the fact that the intervention changes exactly the corridor under dispute. It does not show that stop-gradient is the best practical method; it shows that the CE corridor is sufficient to destroy the conditioning signature when it dominates.

Gradient dominance should be measured under the matched optimizer. The gradient-competition measurements are taken during Adam training at learning rate 10^{-3} , while the conditioning sweep in the causal experiment uses SGD. The dominance ratio is expected to reflect the loss geometry at the distance interface rather than an Adam-specific artifact, but the stronger version of the result would measure the same CE/EM ratio under the same optimizer and learning-rate sweep used for the conditioning result. Until then, the optimizer mismatch should be stated explicitly.

The optimizer-signature restoration is incomplete. The stop-gradient arm restores SGD learning-rate insensitivity at W_1 , and the ordinary composed arm shows a renewed advantage for Adam. The complementary Adam stop-gradient arm has not yet been reported. That run would test the remaining half of the optimizer-signature claim: whether Adam loses its advantage when the local EM objective owns the W_1 update.

Cosine near zero is a weak alignment claim. The measured CE and EM gradients have cosine near zero. In $K = 25$ dimensions, random directions also have cosine standard deviation of about $1/\sqrt{25} = 0.2$. The correct conclusion is therefore not strong geometric orthogonality. The conservative conclusion is that there is no detectable alignment or opposition at the measured scale. This still supports the mechanism: the EM signal is diluted by a larger foreign signal, not cancelled by a directly opposing one. The observed values are smaller than this coarse random-direction scale, but batch averaging and the shared NLS interface change the relevant null; we do not interpret that undershoot further.

The graded claim is currently supported by few points. The causal experiment includes $\lambda = 0.001$, $\lambda = 0.03$, $\lambda = 1$, and stop-gradient. This is enough to show that the effect is not binary: $\lambda = 0.03$ lies between the CE-dominated and EM-dominated regimes, and $\lambda = 1$ behaves like stop-gradient despite full connectivity. A denser λ -resolution curve, plotted against the measured CE/EM gradient ratio, would make the graded claim much stronger and is the natural next analysis for this result.

The learning-rate sweeps mainly show low-rate slowness. The reported sweeps show large final-accuracy spreads across learning rates, but the SGD arms mostly improve monotonically over

the tested range. The current spread metric therefore primarily captures undertraining at small learning rates, not an observed upper instability edge. A larger learning-rate rung or a time-to-threshold analysis would sharpen the smoothness interpretation.

Some structural metrics are close to their training losses. Dead-unit and redundancy measurements are related to the variance and decorrelation penalties used in the auxiliary objective. This does not make the repair trivial: the cross-metrics matter, for example the variance-only arm removes dead units while making redundancy much worse. Still, the overlap between objective terms and reported health metrics should be read as a targeted repair test rather than as an independent representation-quality measure.

The auxiliary objective has escape directions. The curvature propositions apply to the LSE contribution at the local responsibility site. The full auxiliary objective also includes variance and decorrelation terms, and high-learning-rate Adam can drive the variance term to very large values without improving task accuracy. The reported operating point is chosen below this runaway regime; the theory should not be read as a global smoothness proof for the full auxiliary objective.

The basin interpretation remains incomplete. The discussion interprets discriminative fine-tuning as a dominant foreign gradient that can move the model away from local EM structure. The present evidence supports this as a mechanism-level interpretation, but it does not establish that EM-dominated and CE-dominated solutions occupy different basins. A linear-mode-connectivity analysis with hidden-unit alignment is needed before making a stronger basin claim. Without alignment, apparent barriers could be artifacts of component permutation.

10.2 Limitations of Scope

Dataset and scale. The experiments use MNIST and small supervised networks. This is appropriate for isolating the mechanism, but it limits external validity. We do not claim state-of-the-art performance, broad deep-learning optimization coverage, or automatic transfer to modern large-scale architectures. The claim is that the controlled setting exposes a locality boundary that larger systems must respect when the same structural conditions are present.

Single main architecture. The experimental architecture has one intermediate EM site followed by a dense supervised head. This is the architecture needed to test the corridor hypothesis directly. Other architectures may include normalization, residual paths, attention-style stochastic maps, convolutional sharing, or multiple local sites. The theory predicts which operations should preserve or destroy the simplex structure, but those variants require separate experiments.

The formal results are sufficient conditions, not a characterization. The private-parameter propositions show that local LSE sites can inherit a parameter-independent curvature bound. The composition proposition shows that a learned dense map does not preserve that guarantee. These are not universal necessity claims. Some composed systems may remain well conditioned for unrelated reasons, and some private systems may fail because of effects outside the propositions, such as hard gates or degenerate data geometry.

The theory does not prove composed curvature is always large. Proposition 3 states a loss of guarantee. It shows that the $1/2$ LSE bound no longer controls the composed CE corridor because the bound can depend on the learned spectrum of W_2 . The empirical measurements show that the composed curvature is large in the studied network. The theorem alone does not imply that every dense composition has large curvature at every point.

LayerNorm is handled empirically, not in the propositions. The measured models include LayerNorm where specified, so the reported conditioning and curvature results include its effect. The propositions omit LayerNorm to isolate the LSE and dense-corridor geometry. A fully expanded proof would insert the LayerNorm Jacobian and additional input-dependent terms. This would complicate the constants but would not restore the parameter-independent LSE guarantee through a learned dense corridor.

No convergence theorem is provided for the composed system. The paper analyzes local curvature and measures optimization signatures, but it does not prove convergence of the full supervised system. The results should be read as a conditioning and mechanism study, not as a global optimization theory for neural networks.

Future strengthening analyses. A denser λ -resolution curve, plotted against the measured CE/EM gradient ratio, would strengthen the graded claim. A basin or linear-mode-connectivity analysis with hidden-unit alignment would test whether EM-dominated and CE-dominated checkpoints occupy distinct basins. A nonnegative row-normalized corridor would test the simplex-compatible preservation class separately from the dense-corridor arms. These analyses are useful extensions; the current volume-control, stop-gradient, dominance, and curvature results do not depend on them.

The stochastic-map bridge does not carry the main claims. A row-normalized nonnegative corridor is an important preservation-class test and a useful bridge to attention, but the main claims do not rest on it. If the stochastic-map arm is unconfounded, it strengthens the operation taxonomy. If it is confounded by reduced task accuracy or head capacity, it should remain an appendix-level result. The core claims are already tested by the volume-control ablations, optimizer sweep, stop-gradient intervention, gradient dominance, and curvature measurements.

11 Conclusion

Log-sum-exp objectives make responsibilities appear as gradients. This gives standard neural objectives a precise implicit-EM interpretation, but the interpretation has a boundary. It is a statement about the coordinates seen by an LSE or softmax objective, not a guarantee that a whole composed network inherits EM’s optimizer geometry.

The theory identifies that boundary. In distance coordinates, the local LSE Hessian is uniformly bounded by $1/2$. When component parameters are private and affine, this yields a parameter-independent curvature bound and explains the single-site optimizer signature observed in earlier work. When the same site is trained through a learned dense corridor, the leading term of the composed CE Hessian is a conjugated softmax Hessian whose scale depends on the learned map. The local guarantee is therefore lost.

The supervised experiments bear out the split in this controlled setting: volume control transfers to an intermediate EM site, but the single-site optimizer conditioning does not survive ordinary composition—and is restored, causally, by cutting the corridor or letting the EM path dominate.

The resulting rule is local, graded, and structural. EM conditioning is visible when the responsibility-gradient path directly controls the relevant parameters or when the EM path dominates the effective update; ordinary dense backpropagation should not be expected to preserve it. Simplex-compatible transport remains a theoretical first-order preservation class, not yet an empirically tested one.

Two consequences point beyond this paper. For architecture, the result motivates building depth as protected local EM sites rather than expecting a single owned objective to condition every layer. For interpretation, it says the intermediate softmax sites of standard attention architectures, owning no objective of their own, should not be read as automatically inheriting EM’s optimization structure. Knowing which properties transfer, which fail, and which interventions make them survive is the boundary the implicit-EM framework now provides.

A Notation and Conventions

This appendix collects the sign convention and notation used throughout the paper. We use *energy* and *distance* interchangeably; lower values indicate better matches. For an input x and K components, d_j denotes the energy of component j . Responsibilities are defined by

$$r = \text{softmax}(-d), \quad r_j = \frac{\exp(-d_j)}{\sum_k \exp(-d_k)},$$

so that $r_j \geq 0$ and $\sum_j r_j = 1$. In this convention, the log-sum-exp objective is the negative log marginal

$$L_{\text{LSE}}(d) = -\log \sum_j \exp(-d_j),$$

and differentiating with respect to a distance gives

$$\frac{\partial L_{\text{LSE}}}{\partial d_j} = r_j.$$

The sign is positive because increasing a distance increases the negative log marginal. Writing logits or negative energies instead yields the same identity with the opposite sign. All results in the paper use the distance convention.

For supervised classification, $h \in \mathbb{R}^C$ denotes class logits and

$$p = \text{softmax}(h)$$

denotes the output-layer class posterior. The label-clamped cross-entropy gradient is

$$\frac{\partial L_{\text{CE}}}{\partial h_j} = p_j - \mathbb{1}\{j = y\}.$$

We reserve r for component responsibilities at an intermediate EM site and p for output class probabilities. We write $\mathbb{1}\{\cdot\}$ for the indicator function and $\mathbf{1}$ for the all-ones vector.

The supervised model contains two maps of interest. The first, W_1 , sends the input to component energies; its rows are treated as private component parameters in the single-site theory. The second, W_2 , sends component coordinates to class logits. It is the learned dense component-to-class

map, or *corridor*, through which the supervised gradient is pulled back. Its largest singular value is written $\sigma_{\max}(W_2)$.

The NegLogSoftmin calibration used in the experiments is

$$\text{NLS}(d)_j = d_j + \log \sum_k \exp(-d_k).$$

It preserves pairwise distance differences and satisfies

$$\exp(-\text{NLS}(d)_j) = r_j,$$

converting raw distances into calibrated distances whose exponentials are responsibilities.

Volume control consists of an anti-collapse variance penalty L_{var} and an anti-redundancy decorrelation (total-correlation) penalty L_{tc} . We write

$$L_{\text{aux}}(d) = L_{\text{LSE}}(d) + L_{\text{var}}(d) + L_{\text{tc}}(d)$$

for the local auxiliary EM objective with all terms active. Unless otherwise stated, λ denotes the total auxiliary weight in the supervised objective,

$$L_{\text{total}} = L_{\text{CE}} + \lambda L_{\text{aux}}.$$

(The accompanying experiment scripts refer to this quantity as `lambda_reg`.)

Finally, we use the spectral norm of the LSE Hessian in distance coordinates,

$$\|H_d L_{\text{LSE}}\|_2,$$

as a site-level sharpness diagnostic. Since the LSE Hessian is bounded above by $1/2$ [6], we refer to proximity to this bound as the *1/2-saturation diagnostic*. Saturation indicates sharp two-way component competition at the site; values well below $1/2$ indicate diffuse responsibilities.

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Test	Prediction	Measured	Status
Exp1	Supervision does not give intermediate volume control	Baseline has min variance zero and ≥ 1 dead unit in every seed	Supported
Exp1	Partial volume control is worse than none	Redundancy increases from 20.5 to 187.0; accuracy is worst	Supported
Exp3	Optimizer signature disappears under composition	SGD spans 83.8–96.2%; Adam helps	Supported
Exp4-P1	CE dominates EM at small λ	Late CE/EM ratio is 67–95 $\times$ at $\lambda = 0.001$	Supported
Exp4-P2	LSE curvature stays below 1/2; CE-corridor curvature is larger	LSE bound holds; CE corridor is 13–100 $\times$ larger	Supported
Exp4-P3	Stop-gradient restores conditioning	Learning-rate spread falls from 10.9 to 1.6 points	Supported
Exp4	$\lambda = 1$ joint behaves like stop-gradient (dominance, not connectivity)	Spread 1.2 vs 1.6 points; both are nearly flat	Supported
Exp4	Restored conditioning costs task alignment	EM-dominated features probe $\sim 88\%$ vs $\sim 96\%$	Supported
Exp4	CE/EM dominance should hold under the matched SGD sweep	Gradient ratio currently measured during Adam at 10^{-3}	Open
Exp4	Adam stop-gradient should lose the ordinary Adam advantage	Stop-gradient sweep currently reported for SGD	Open
Theory/Exp4	Curvature scale should be compared to the $\mathbb{E}\ \tilde{x}\ ^2$ prediction	For MNIST, $\mathbb{E}\ \tilde{x}\ ^2 = 88.8$, giving an LSE descent scale of $2/L \approx 0.045$; stop-gradient is stable through $\eta = 0.1$	Reported with caveat

Table 5: Prediction ledger for the supervised study. The table separates mechanism predictions, measurements, status, and post-measurement interpretation. “Supported” means supported by the reported measurements, not mathematically closed under every strengthening check. “Reported with caveat” marks a quantitative scale check whose order of magnitude is reported but whose idealized assumptions do not exactly match the experimental arm. One interpretation was corrected after measurement: the early heuristic claim that $\lambda = 0.001$ implied a 1000 $\times$ gradient ratio was wrong; the measured late-training ratio is 67–95 $\times$ (Section 8, gradient competition). The full trajectory is wider because of early transients. The mechanism is unchanged; the magnitude was corrected.

Arm	Objective	CE reaches W_1 ?
Joint, $\lambda = 0.001$	$L_{\text{CE}} + 0.001L_{\text{aux}}$	yes
Joint, $\lambda = 0.03$	$L_{\text{CE}} + 0.03L_{\text{aux}}$	yes
Joint, $\lambda = 1$	$L_{\text{CE}} + L_{\text{aux}}$	yes
Stop-gradient	$L_{\text{CE}}(\text{head}(\text{stopgrad}(y))) + L_{\text{aux}}$	no

Table 6: Arms for the causal experiment. The stop-gradient arm leaves the supervised head, labels, data, and architecture in place, but blocks the CE gradient from updating W_1 .

Arm	10^{-4}	10^{-3}	10^{-2}	10^{-1}	Spread
Joint, $\lambda = 0.001$	85.30 ± 2.53	92.99 ± 0.15	95.05 ± 0.06	96.19 ± 0.17	10.9
Joint, $\lambda = 0.03$	86.42 ± 0.28	90.53 ± 0.04	91.78 ± 0.13	92.23 ± 0.29	5.8
Joint, $\lambda = 1$	88.35 ± 0.07	88.20 ± 0.10	88.41 ± 0.09	87.26 ± 0.52	1.2
Stop-gradient	88.37 ± 0.08	88.10 ± 0.07	87.94 ± 0.16	86.73 ± 0.50	1.6

Table 7: Linear-probe accuracy on frozen intermediate distances. Values are mean $\pm$ standard deviation over 3 seeds, in percent.

Arm	10^{-4}	10^{-3}	10^{-2}	10^{-1}
Stop-gradient	32.0	27.7	21.7	23.7
Joint, $\lambda = 0.001$	32.3	15.0	40.0	37.7

Table 8: Epoch when test L_{LSE} reaches 95% of its final value. The two extreme arms (CE-blocked local EM path vs. strongly CE-dominated) are shown because they bound the contrast; this metric was not recorded for the intermediate $\lambda = 0.03$ and $\lambda = 1$ arms. The CE-dominated arm should be read as an EM-site stability contrast, not as a classifier failure metric. Because the threshold is normalized to each run’s own final value, the table is a self-normalized timing diagnostic rather than an absolute convergence threshold.

Arm	Early CE/EM ratio	Late CE/EM ratio	Cosine
Joint, $\lambda = 0.001$	0.18–0.74×	67–95×	−0.05 to 0.05
Joint, $\lambda = 1$	0.00018–0.00074×	0.018–0.025×	−0.07 to 0.08

Table 9: Gradient competition at the intermediate distance coordinates. The ratio uses the weighted auxiliary gradient, $\lambda \|\nabla_d L_{\text{aux}}\|$. Early values are the first recorded measurement for each seed; late values are the final recorded measurement.